

But the advantages of blockchain need to go way beyond traceability and security. It should build trust, and support frontline staff in planning nursing care across clinical pathways, making sure outcomes are recorded, so that the administrative workload of nurses can be reduced significantly. The blockchain should give the same information to nurses about a patient, irrespective of where the former are located, in order to ensure continuity of the health service.

Optimisation of clinical pathways through blockchain and artificial intelligence (AI) will help to store nurses' activities, as well as monitor the data that is continuously fed into the nursing care system so that they can plan daily activities better. A bidirectional flow of data between patients and authorised caretakers makes data accessible across healthcare networks, which is needed to make healthcare effective.

Some interesting initiatives have been launched to manage patients who suffer from chronic health issues, which combine distributed ledger technologies (DLT) with novel medical IoT devices (IoMT) to facilitate patient remote management and monitoring, as well as self-management. The adoption of IoMT can help improve the quality of care and reduce the occurrence of severe events, which might otherwise lead to hospitalisation, while at the same time reducing unnecessary visits to emergency rooms. The data coming from these devices needs to meet basic quality and safety criteria, which DLTs can help achieve.

Use cases of blockchain in the healthcare industry

Since blockchain is immutable, distributed and decentralised, it is being explored for various use cases in the healthcare and medical services industries. As per Gartner, the current pandemic has catalysed the evolution of applications that can be built on

blockchain for use in health services:

- It offers liquidity to expedite business transactions (such as purchase of medical devices) in case of a financial emergency.
- Improves trust in supply chain processing and helps track assets with high transparency.
- Digital document signing helps with processing of insurance claims.
- Enables zero touch payments.
- Cryptocurrencies can be used for supply chain activities.

Online shopping for medicines

and medical supplies: In one of its Top 10 strategic predictions, Gartner Hype Cycle (a graphical representation of the perceived value of a technology trend or innovation) indicates that there has been at least a 30 per cent increase in blockchain solutions for healthcare.

Online shopping for medical supplies and medicines has increased a lot in the pandemic and requires validation of quantity, medicine type, permissions for drug consumption, etc. Hence, an approval process is inevitable to process such orders. In a blockchain platform, multiple nodes act as approvers and the transactions can be easily processed with the help of proof of work (PoW) or proof of stake (PoS) algorithms.

Active patient monitoring:

Remote patient monitoring is a sought-after service in the healthcare industry. IoT enabled sensors can be implanted in patients for taking vital statistics like body temperature, oxygen level, pulse rate, heart rate and that information can be transmitted to remote hospitals.

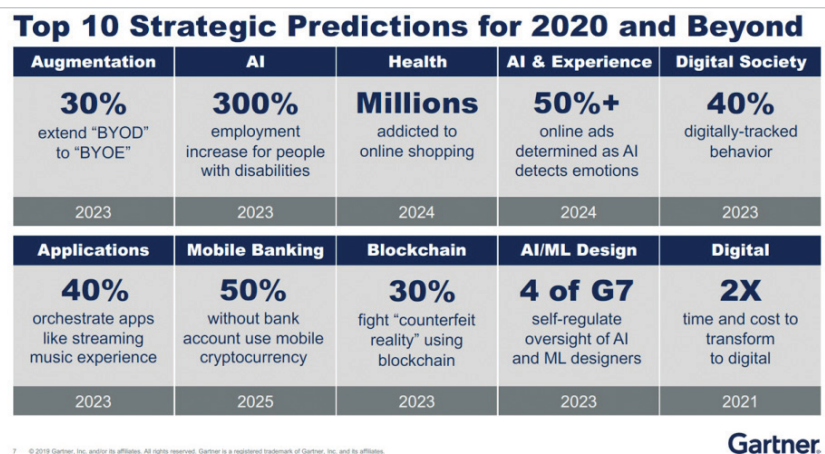


Figure 1: Strategic predictions for 2020 as per Gartner

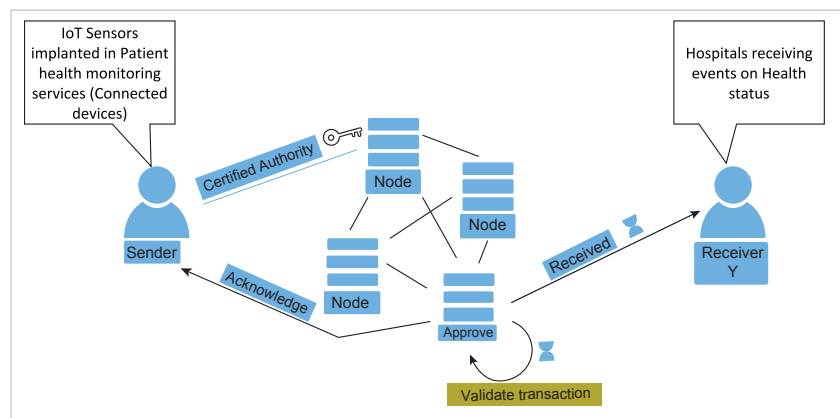


Figure 2: Active patient monitoring using blockchain platform